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PROJECT REVIEW - 1

TITLE: Explainable AI Network Intrusion Detection Systems (NIDS)

COURSE NAME : PYTHON PROGRAMMING

COURSE TYPE : ETH

COURSE CODE : PAAML503

SLOT : C1+TC1

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Abstract

Network Intrusion Detection Systems (NIDS) play a critical role in modern cybersecurity by continuously monitoring network traffic and identifying malicious or abnormal activities. While machine learning-based NIDS solutions have demonstrated high detection accuracy, their practical adoption is often limited by a lack of transparency in decision-making. Security analysts require not only accurate predictions but also clear explanations of why a network flow is classified as malicious and which traffic characteristics influence that decision. To address this challenge, this project proposes an Explainable Artificial Intelligence (XAI)-based flow-level Network Intrusion Detection System using the CICIDS2017 dataset.

Among the widely used intrusion detection datasets, CIC-UNSW-NB15, CICIDS2017, and CSE-CIC-IDS2018 were evaluated as potential data sources. CICIDS2017 was selected as the primary dataset because it provides realistic network traffic, comprehensive attack scenarios, and more than 80 flow-level statistical features generated through CICFlowMeter. The dataset contains both benign traffic and diverse cyberattacks, including Brute Force, Denial of Service (DoS), Distributed Denial of Service (DDoS), PortScan, Web Attacks, Botnet activity, Infiltration, and Heartbleed attacks. Using a single dataset ensures consistency in network topology, feature representation, and labeling, thereby avoiding biases introduced by combining heterogeneous datasets.

The proposed methodology begins with data cleaning and preprocessing, including the removal of duplicate records, handling missing and invalid values, standardizing labels, and eliminating features that may introduce data leakage. Exploratory Data Analysis (EDA) is conducted to examine class distributions, feature correlations, outliers, and traffic behavior patterns. Additional feature engineering techniques are applied to derive meaningful traffic indicators such as packet rates, byte rates, traffic asymmetry measures, and inter-arrival time statistics.

A hierarchical classification framework is adopted in which the first stage performs binary classification to distinguish benign and malicious traffic, while the second stage categorizes malicious traffic into specific attack families. Multiple machine learning models, including XGBoost, Random Forest, and Multi-Layer Perceptron (MLP), are employed as base learners. Their outputs are integrated through a stacking ensemble architecture to improve detection performance and robustness. Furthermore, a Genetic Algorithm (GA) is incorporated for feature selection, reducing dimensionality and improving model interpretability by identifying the

most informative network-flow features.

To enhance transparency and trustworthiness, the project integrates Explainable AI techniques, primarily SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). SHAP provides both global and local explanations by quantifying the contribution of individual features to model predictions, while LIME offers intuitive explanations for specific network alerts and challenging classification cases. The combined use of these techniques enables analysts to understand not only whether a network flow is malicious but also the underlying factors driving the model's decision.

The proposed system aims to deliver a reliable, interpretable, and scalable intrusion detection framework that balances predictive performance with explainability. By combining advanced machine learning, feature optimization, and XAI methodologies, the project contributes toward the development of transparent cybersecurity solutions capable of supporting informed decision-making in real-world network environments.

